



AMRITA
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22MAT122 - MATHEMATICS FOR COMPUTING - 2

DEEP CONVOLUTION TABLE : DEEP LEARNING WITHOUT CONVOLUTIONS

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Step 1: Load CIFAR-10 Dataset

This CT model uses the dataset CIFAR-10 which has 60,000 32x32 colour images in 10 classes, with 6000 images per class. There are 50000 training images and 10000 test images.

- Each image is 32×32 pixels with 3 color channels (RGB).
- Categorical() is required because the training function expects class labels in that format.

```
% The CIFAR-10 dataset consists of 60,000 color images (32x32 pixels) in 10 classes.  
% We split this into 50,000 training images and 10,000 test images.
```

```
% Datasetdisp('Loading CIFAR-10 dataset...');
```

```

clc; clear; close all;
data_path = 'C:\Users\sandh\OneDrive\Desktop\projects-sem 2\EOC+MFC\data\cifar-10-
batches-mat\';

[trainImages, trainLabels, testImages, testLabels] = loadCIFAR10Data(data_path);

if isempty(trainImages) || isempty(testImages)
    error('Failed to load CIFAR-10 data. Check data path or file integrity.');
```

end

Step 2: Preprocessing

We are normalizing the values because the neural network converges faster when the input values are normalised.

```

% Normalize images by scaling RGB values from [0, 255] → [0, 1]
% Subtract mean image for zero-centering

fprintf('Step 2: Preprocessing data...\n');
```

Step 2: Preprocessing data...

```

trainImages = im2double(trainImages);
testImages = im2double(testImages);

% Calculate mean of training images to normalize both train and test sets

meanImage = mean(trainImages, 4);
trainImages = trainImages - meanImage;
testImages = testImages - meanImage;

% Convert labels to categorical format for training

trainLabels = categorical(trainLabels);
testLabels = categorical(testLabels);
```

Step 3: Data Augmentation

We use techniques like random rotation, horizontal flipping, scaling and translation to increase the diversity of training images. This helps the model generalize better by making it more robust to many variations.

```

% Helps improve generalization by introducing image variations like:
% - Random rotation
% - Horizontal flipping
% - Scaling and translation
% It also prevents data overfitting

fprintf('Step 3: Creating augmented datastore...\n');
```

Step 3: Creating augmented datastore...

```

imageSize = [32 32 3];

augmter = imageDataAugmenter( ...
    'RandRotation', [-10, 10], ...
    'RandXTranslation', [-3 3], ...
    'RandYTranslation', [-3 3], ...
    'RandXReflection', true);

augimdsTrain = augmentedImageDatastore(imageSize, trainImages, trainLabels,
    'DataAugmentation', augmter);
augimdsTest = augmentedImageDatastore(imageSize, testImages, testLabels);

```

Step 4: Load Pretrained ResNet-18 and Modify

ResNet-18 is pretrained on ImageNet (224 x 224 images, 1000 classes)

$$T^{\text{in}} \in \mathbb{R}^{H_i \times W_i \times D_i}$$

% We use 224 x 224 as input because ResNet-18 expects images of the particular size as input. Also, we add new layers for 10-class classification (CIFAR-10).

```
fprintf('Step 4: Loading and modifying ResNet-18...\n');
```

Step 4: Loading and modifying ResNet-18...

```

net = resnet18;
lgraph = layerGraph(net);

% Modify the input size to match CIFAR-10's input size
lgraph = replaceLayer(lgraph, 'data', imageInputLayer([32 32 3], 'Name', 'input',
    'Normalization', 'none'));

% Replace final layers for 10-class classification
numClasses = 10;
lgraph = replaceLayer(lgraph, 'fc1000', fullyConnectedLayer(numClasses, 'Name',
    'fc10'));
lgraph = replaceLayer(lgraph, 'ClassificationLayer_predictions',
    classificationLayer('Name', 'output'));

```

Step 5: Training Options

SGDM (Stochastic Gradient Descent with Momentum) is used again.

Learning Rate Schedule: piecewise drops the learning rate by DropFactor after DropPeriod epochs.

$$\alpha_{\text{new}} = \alpha_{\text{old}} \times \text{DropFactor}$$

(eg. 0.01 --> 0.001 after 30 epochs)

The training runs on CPU.

-> We use the Stochastic Gradient Descent method in the convolution tables method with momentum to help avoid the local minima by :

$$v_t = \mu v_{t-1} - \eta \nabla L(\theta)$$

$$\theta = \theta + v_t$$

where μ is the momentum and η is the learning rate.

```
fprintf('Step 5: Setting training options...\n');
```

Step 5: Setting training options...

```
options = trainingOptions('sgdm', ...
    'InitialLearnRate', 0.01, ...
    'MaxEpochs', 30, ...
    'MiniBatchSize', 128, ...
    'Shuffle', 'every-epoch', ...
    'ValidationData', augimdsTest, ...
    'ValidationFrequency', 100, ...
    'Verbose', true, ...
    'Plots', 'training-progress', ...
    'ExecutionEnvironment', 'cpu', ...
    'LearnRateSchedule', 'piecewise', ...
    'LearnRateDropFactor', 0.1, ...
    'LearnRateDropPeriod', 30);
```

Step 6: Train the Model

Backpropagation runs here to minimize the loss.

Training uses the specified options, including learning rate schedule and data augmentation.

```
fprintf('Step 6: Training ResNet-18...\n');
```

Step 6: Training ResNet-18...

```
net = trainNetwork(augimdsTrain, lgraph, options);
```

Epoch	Iteration	Time Elapsed (hh:mm:ss)	Mini-batch Accuracy	Validation Accuracy	Mini-batch Loss	Validation Loss	Base Learning Rate
1	1	00:02:08	10.16%	14.43%	3.7786	3.2924	0.01
1	50	00:04:22	44.53%		1.6565		0.01
1	100	00:08:14	59.38%	61.60%	1.2308	1.1078	0.01
1	150	00:10:20	61.72%		1.0530		0.01
1	200	00:13:39	55.47%	66.38%	1.1909	0.9526	0.01
1	250	00:14:41	67.19%		0.9437		0.01
1	300	00:16:33	64.84%	71.57%	0.9860	0.8297	0.01
1	350	00:17:27	75.00%		0.7732		0.01

2	400	00:18:57	73.44%	74.96%	0.7820	0.7336	0.03
2	450	00:20:00	67.97%		0.9044		0.03
2	500	00:21:43	74.22%	75.42%	0.7000	0.7067	0.03
2	550	00:22:29	70.31%		0.9613		0.03
2	600	00:24:22	72.66%	76.04%	0.7872	0.7090	0.03
2	650	00:25:23	73.44%		0.7585		0.03
2	700	00:27:17	82.81%	77.39%	0.5733	0.6530	0.03
2	750	00:28:18	75.00%		0.7151		0.03
3	800	00:30:12	78.91%	78.51%	0.5796	0.6295	0.03
3	850	00:31:14	80.47%		0.5806		0.03
3	900	00:33:06	78.91%	78.51%	0.6246	0.6312	0.03
3	950	00:34:07	69.53%		0.8711		0.03
3	1000	00:36:02	77.34%	78.07%	0.8892	0.6449	0.03
3	1050	00:37:04	74.22%		0.7212		0.03
3	1100	00:38:58	82.03%	79.56%	0.5002	0.6052	0.03
3	1150	00:39:59	78.91%		0.6488		0.03
4	1200	00:41:53	81.25%	80.06%	0.5931	0.5899	0.03
4	1250	00:42:54	79.69%		0.4922		0.03
4	1300	00:44:48	80.47%	80.42%	0.5437	0.5774	0.03
4	1350	00:45:52	81.25%		0.5813		0.03
4	1400	00:47:49	78.91%	80.65%	0.5961	0.5751	0.03
4	1450	00:48:51	73.44%		0.6941		0.03
4	1500	00:50:18	74.22%	80.90%	0.7191	0.5622	0.03
4	1550	00:50:44	84.38%		0.4866		0.03
5	1600	00:52:18	79.69%	80.32%	0.4866	0.5704	0.03
5	1650	00:53:20	80.47%		0.5857		0.03
5	1700	00:55:32	82.03%	80.24%	0.5075	0.5854	0.03
5	1750	00:56:38	83.59%		0.5008		0.03
5	1800	00:58:28	81.25%	80.20%	0.4910	0.5727	0.03
5	1850	00:59:42	82.03%		0.5886		0.03
5	1900	01:01:37	81.25%	81.21%	0.5183	0.5578	0.03
5	1950	01:02:41	81.25%		0.6767		0.03
6	2000	01:04:41	82.03%	80.88%	0.6643	0.5794	0.03
6	2050	01:05:44	85.16%		0.4456		0.03
6	2100	01:07:42	82.81%	81.98%	0.4887	0.5351	0.03
6	2150	01:08:45	87.50%		0.3893		0.03
6	2200	01:10:27	77.34%	81.92%	0.5344	0.5351	0.03
6	2250	01:11:31	78.91%		0.5320		0.03
6	2300	01:13:30	83.59%	81.25%	0.5381	0.5456	0.03
7	2350	01:14:35	82.81%		0.4558		0.03
7	2400	01:16:34	85.94%	80.89%	0.3755	0.5646	0.03
7	2450	01:17:38	79.69%		0.7221		0.03
7	2500	01:19:36	82.81%	82.67%	0.4775	0.5195	0.03
7	2550	01:20:41	84.38%		0.5257		0.03
7	2600	01:22:39	76.56%	82.73%	0.6882	0.5174	0.03
7	2650	01:23:43	82.03%		0.4787		0.03
7	2700	01:25:41	78.12%	82.88%	0.5730	0.5110	0.03
8	2750	01:26:46	83.59%		0.4955		0.03
8	2800	01:28:45	83.59%	82.66%	0.4599	0.5061	0.03
8	2850	01:29:49	78.12%		0.6223		0.03
8	2900	01:31:45	86.72%	81.24%	0.4495	0.5487	0.03
8	2950	01:32:49	84.38%		0.4957		0.03
8	3000	01:34:47	85.94%	81.86%	0.4051	0.5305	0.03
8	3050	01:35:52	78.12%		0.6704		0.03
8	3100	01:37:51	80.47%	81.75%	0.6468	0.5464	0.03
9	3150	01:38:57	85.94%		0.3790		0.03
9	3200	01:40:55	91.41%	82.71%	0.3351	0.5168	0.03
9	3250	01:42:53	78.91%		0.5963		0.03
9	3300	01:46:35	84.38%	83.02%	0.4333	0.5016	0.03
9	3350	01:48:36	80.47%		0.4942		0.03
9	3400	01:52:18	88.28%	83.30%	0.2958	0.5006	0.03
9	3450	01:54:18	82.03%		0.4554		0.03
9	3500	01:57:59	82.81%	82.44%	0.4873	0.5193	0.03
10	3550	02:00:02	85.94%		0.4011		0.03

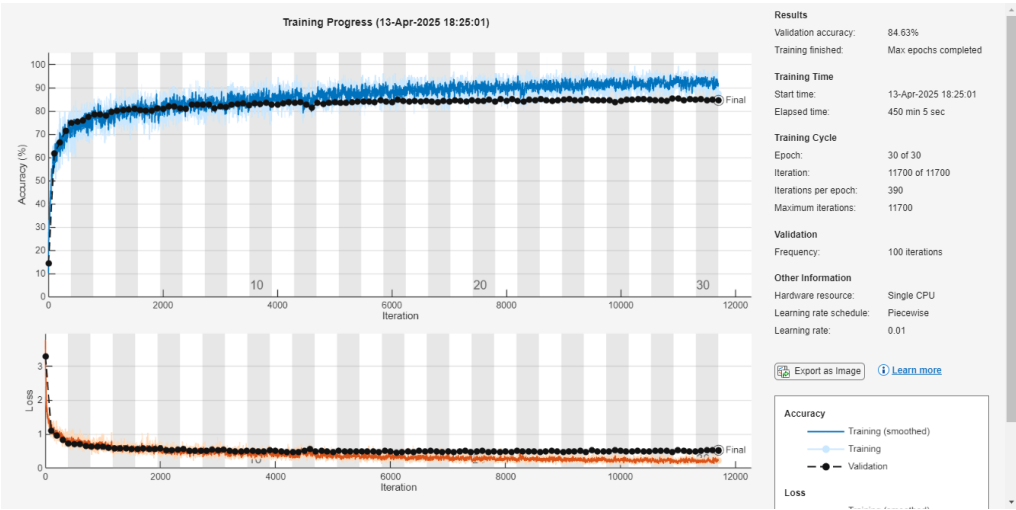
10	3600	02:03:42	86.72%	83.39%	0.4029	0.5070	0.03
10	3650	02:05:39	88.28%		0.3709		0.03
10	3700	02:09:12	82.03%	83.14%	0.5585	0.4941	0.03
10	3750	02:11:14	89.06%		0.3282		0.03
10	3800	02:14:57	83.59%	83.51%	0.3807	0.4899	0.03
10	3850	02:16:58	84.38%		0.4347		0.03
10	3900	02:20:41	79.69%	82.76%	0.5923	0.5261	0.03
11	3950	02:22:45	86.72%		0.3811		0.03
11	4000	02:26:24	84.38%	82.82%	0.3877	0.5137	0.03
11	4050	02:28:24	87.50%		0.4204		0.03
11	4100	02:32:04	81.25%	83.37%	0.4270	0.4979	0.03
11	4150	02:34:04	86.72%		0.4441		0.03
11	4200	02:37:45	89.06%	83.87%	0.3736	0.4833	0.03
11	4250	02:39:45	85.94%		0.4310		0.03
12	4300	02:43:21	87.50%	83.68%	0.3364	0.4789	0.03
12	4350	02:45:29	88.28%		0.3692		0.03
12	4400	02:49:13	88.28%	83.93%	0.3787	0.4815	0.03
12	4450	02:51:16	83.59%		0.4742		0.03
12	4500	02:55:01	89.84%	82.81%	0.3745	0.5156	0.03
12	4550	02:57:04	82.81%		0.4738		0.03
12	4600	03:00:53	85.94%	81.45%	0.4452	0.5573	0.03
12	4650	03:02:58	82.81%		0.5343		0.03
13	4700	03:06:47	88.28%	83.67%	0.3700	0.4900	0.03
13	4750	03:08:49	85.16%		0.3445		0.03
13	4800	03:12:34	82.81%	83.09%	0.5613	0.5120	0.03
13	4850	03:14:38	88.28%		0.3586		0.03
13	4900	03:17:28	87.50%	83.61%	0.3766	0.4876	0.03
13	4950	03:18:28	89.06%		0.3590		0.03
13	5000	03:20:18	85.16%	83.87%	0.4305	0.4842	0.03
13	5050	03:21:17	84.38%		0.4120		0.03
14	5100	03:23:07	88.28%	83.67%	0.3276	0.5025	0.03
14	5150	03:24:06	87.50%		0.3706		0.03
14	5200	03:25:54	85.94%	83.48%	0.2997	0.4973	0.03
14	5250	03:26:53	82.81%		0.5262		0.03
14	5300	03:28:42	89.06%	83.87%	0.2969	0.4966	0.03
14	5350	03:29:41	91.41%		0.2396		0.03
14	5400	03:31:32	86.72%	84.15%	0.3840	0.4851	0.03
14	5450	03:32:35	84.38%		0.4220		0.03
15	5500	03:34:30	89.06%	84.00%	0.2944	0.4927	0.03
15	5550	03:35:34	87.50%		0.2958		0.03
15	5600	03:37:14	87.50%	84.12%	0.3366	0.4852	0.03
15	5650	03:38:13	89.06%		0.3333		0.03
15	5700	03:40:05	82.03%	83.81%	0.4660	0.4930	0.03
15	5750	03:41:06	86.72%		0.4357		0.03
15	5800	03:42:58	89.84%	84.61%	0.3009	0.4689	0.03
15	5850	03:44:00	86.72%		0.3550		0.03
16	5900	03:45:57	91.41%	84.15%	0.3054	0.5046	0.03
16	5950	03:47:00	86.72%		0.2946		0.03
16	6000	03:48:45	90.62%	83.82%	0.2390	0.4865	0.03
16	6050	03:49:12	87.50%		0.3989		0.03
16	6100	03:50:02	89.06%	84.74%	0.3285	0.4650	0.03
16	6150	03:50:29	90.62%		0.3091		0.03
16	6200	03:52:02	89.84%	84.24%	0.2802	0.4739	0.03
17	6250	03:53:07	89.06%		0.3435		0.03
17	6300	03:55:05	95.31%	84.14%	0.2124	0.4977	0.03
17	6350	03:56:10	86.72%		0.3316		0.03
17	6400	03:58:07	90.62%	84.28%	0.3186	0.4795	0.03
17	6450	03:59:11	84.38%		0.3666		0.03
17	6500	04:01:09	87.50%	84.13%	0.3549	0.5046	0.03
17	6550	04:02:11	87.50%		0.3817		0.03
17	6600	04:04:05	91.41%	84.34%	0.3345	0.4928	0.03
18	6650	04:05:09	84.38%		0.4600		0.03
18	6700	04:07:04	91.41%	84.01%	0.2558	0.5049	0.03
18	6750	04:08:07	92.19%		0.1996		0.03

18	6800	04:10:03	92.97%	83.84%	0.2364	0.4946	0.03
18	6850	04:10:44	88.28%		0.3187		0.03
18	6900	04:12:40	87.50%	84.32%	0.3577	0.4740	0.03
18	6950	04:13:43	88.28%		0.3382		0.03
18	7000	04:15:36	85.94%	84.10%	0.4134	0.4933	0.03
19	7050	04:16:39	90.62%		0.2659		0.03
19	7100	04:18:36	92.19%	84.54%	0.2883	0.4823	0.03
19	7150	04:19:39	94.53%		0.1578		0.03
19	7200	04:21:35	95.31%	84.38%	0.1493	0.4957	0.03
19	7250	04:22:38	89.84%		0.3330		0.03
19	7300	04:24:35	89.84%	84.14%	0.3824	0.5124	0.03
19	7350	04:25:39	89.06%		0.3229		0.03
19	7400	04:27:35	85.94%	84.71%	0.3160	0.4827	0.03
20	7450	04:28:40	85.94%		0.3767		0.03
20	7500	04:30:39	87.50%	84.46%	0.3527	0.4881	0.03
20	7550	04:31:42	89.84%		0.2519		0.03
20	7600	04:33:39	94.53%	84.41%	0.2006	0.5067	0.03
20	7650	04:34:43	85.94%		0.3137		0.03
20	7700	04:36:38	89.84%	84.80%	0.2770	0.4698	0.03
20	7750	04:37:30	85.16%		0.4020		0.03
20	7800	04:39:19	85.16%	84.01%	0.3533	0.4928	0.03
21	7850	04:40:19	88.28%		0.3379		0.03
21	7900	04:42:11	88.28%	85.16%	0.3553	0.4774	0.03
21	7950	04:43:13	89.84%		0.2634		0.03
21	8000	04:45:02	87.50%	84.30%	0.3810	0.4911	0.03
21	8050	04:46:03	86.72%		0.3573		0.03
21	8100	04:47:49	89.06%	85.03%	0.2488	0.4825	0.03
21	8150	04:48:46	92.19%		0.3099		0.03
22	8200	04:50:38	96.88%	84.39%	0.1391	0.5193	0.03
22	8250	04:51:37	87.50%		0.3922		0.03
22	8300	04:53:28	86.72%	84.06%	0.4442	0.4964	0.03
22	8350	04:54:27	89.06%		0.2957		0.03
22	8400	04:56:11	85.94%	84.98%	0.3456	0.4855	0.03
22	8450	04:57:10	83.59%		0.3291		0.03
22	8500	04:58:59	87.50%	84.64%	0.3836	0.4952	0.03
22	8550	04:59:59	90.62%		0.2153		0.03
23	8600	05:03:39	93.75%	85.21%	0.2341	0.4781	0.03
23	8650	05:05:36	92.97%		0.2110		0.03
23	8700	05:09:13	90.62%	84.59%	0.1785	0.4999	0.03
23	8750	05:11:21	92.97%		0.2178		0.03
23	8800	05:15:09	89.84%	84.40%	0.3170	0.5048	0.03
23	8850	05:17:10	89.84%		0.3853		0.03
23	8900	05:20:56	89.84%	84.59%	0.2645	0.4983	0.03
23	8950	05:23:04	94.53%		0.1870		0.03
24	9000	05:26:54	90.62%	84.83%	0.2136	0.4811	0.03
24	9050	05:28:57	88.28%		0.3939		0.03
24	9100	05:32:16	87.50%	85.15%	0.3017	0.4905	0.03
24	9150	05:34:21	92.19%		0.2415		0.03
24	9200	05:38:03	92.19%	84.56%	0.2256	0.4844	0.03
24	9250	05:40:05	89.06%		0.2508		0.03
24	9300	05:43:51	92.19%	84.61%	0.1963	0.5061	0.03
24	9350	05:45:57	90.62%		0.3378		0.03
25	9400	05:49:41	89.84%	85.05%	0.2357	0.4814	0.03
25	9450	05:51:42	93.75%		0.1973		0.03
25	9500	05:55:01	87.50%	84.55%	0.3069	0.5225	0.03
25	9550	05:57:00	89.84%		0.2602		0.03
25	9600	06:00:40	90.62%	84.70%	0.2820	0.4867	0.03
25	9650	06:02:39	91.41%		0.2296		0.03
25	9700	06:06:17	88.28%	84.53%	0.2221	0.4956	0.03
25	9750	06:08:18	95.31%		0.2361		0.03
26	9800	06:11:59	92.19%	84.63%	0.2154	0.4986	0.03
26	9850	06:13:57	91.41%		0.2374		0.03
26	9900	06:17:34	90.62%	83.80%	0.3335	0.5372	0.03
26	9950	06:19:32	94.53%		0.1613		0.03

26	10000	06:23:09	91.41%	84.26%	0.2701	0.5119	0.03
26	10050	06:25:07	88.28%		0.2457		0.03
26	10100	06:28:44	89.84%	84.75%	0.2912	0.4937	0.03
27	10150	06:30:44	95.31%		0.1591		0.03
27	10200	06:34:22	93.75%	84.90%	0.1968	0.4986	0.03
27	10250	06:36:20	90.62%		0.2497		0.03
27	10300	06:39:59	95.31%	85.18%	0.1702	0.5038	0.03
27	10350	06:41:58	92.19%		0.2476		0.03
27	10400	06:45:35	90.62%	85.07%	0.2789	0.4975	0.03
27	10450	06:47:34	87.50%		0.2585		0.03
27	10500	06:51:11	88.28%	84.87%	0.2681	0.4952	0.03
28	10550	06:53:11	88.28%		0.2558		0.03
28	10600	06:56:26	89.06%	84.59%	0.2588	0.5152	0.03
28	10650	06:57:29	94.53%		0.1318		0.03
28	10700	06:59:25	89.84%	84.62%	0.2179	0.5087	0.03
28	10750	07:00:28	93.75%		0.1450		0.03
28	10800	07:02:22	89.84%	84.42%	0.2545	0.5095	0.03
28	10850	07:03:23	92.19%		0.1558		0.03
28	10900	07:05:15	93.75%	85.27%	0.2317	0.4817	0.03
29	10950	07:06:17	89.06%		0.2998		0.03
29	11000	07:08:14	93.75%	85.32%	0.1783	0.5254	0.03
29	11050	07:09:17	90.62%		0.2106		0.03
29	11100	07:11:14	93.75%	84.59%	0.1844	0.5149	0.03
29	11150	07:12:17	94.53%		0.2214		0.03
29	11200	07:14:13	88.28%	85.05%	0.3445	0.4991	0.03
29	11250	07:15:16	89.84%		0.2942		0.03
29	11300	07:17:11	89.84%	84.95%	0.2876	0.4986	0.03
30	11350	07:18:15	91.41%		0.2260		0.03
30	11400	07:20:13	93.75%	85.11%	0.1740	0.5175	0.03
30	11450	07:21:15	96.09%		0.1454		0.03
30	11500	07:23:13	93.75%	84.63%	0.1536	0.5304	0.03
30	11550	07:24:15	89.84%		0.2533		0.03
30	11600	07:26:12	95.31%	84.84%	0.1497	0.5206	0.03
30	11650	07:27:15	88.28%		0.3155		0.03
30	11700	07:29:12	87.50%	84.71%	0.2332	0.5101	0.03

=====

Training finished: Max epochs completed.



Step 7: Evaluate the Model

The **confusion matrix** shows how well the model predicts each class by comparing predicted vs actual labels. Each row represents actual classes, and each column shows predicted ones and the diagonal values indicate correct predictions, while off-diagonal values show misclassifications.

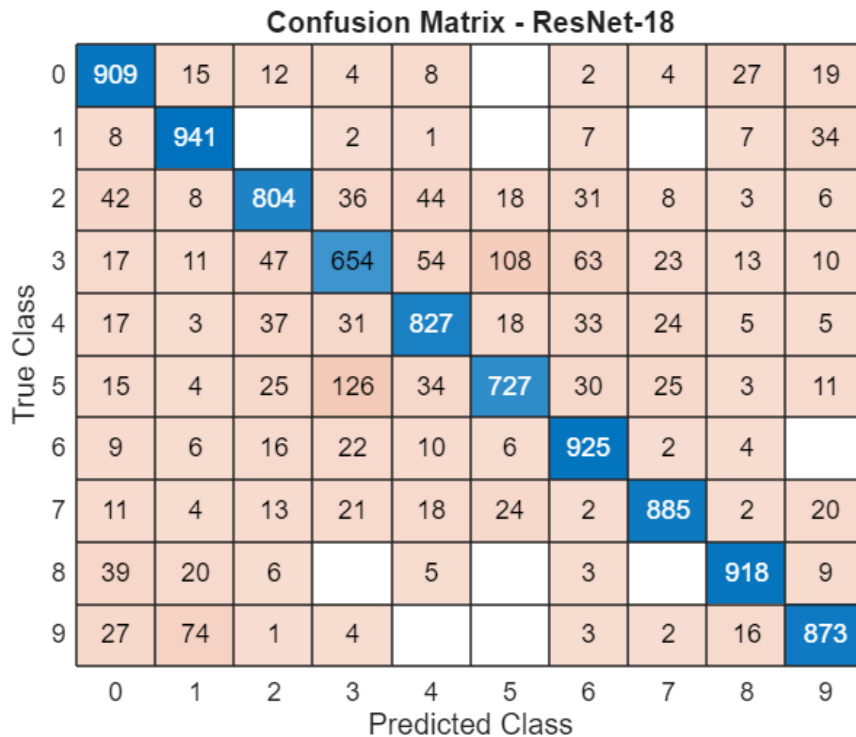

```
fprintf('Step 7: Evaluating test data...\n');
```

Step 7: Evaluating test data...

```
predictedLabels = classify(net, augimdsTest);  
accuracy = sum(predictedLabels == testLabels) / numel(testLabels);  
fprintf('Test Accuracy: %.2f%\n', accuracy * 100);
```

Test Accuracy: 84.63%

```
figure;  
confusionchart(testLabels, predictedLabels);  
title('Confusion Matrix - ResNet-18');
```



Step 8: Save the Accuracy

```
fprintf('Saving Accuracy...\n');
```

Saving Accuracy...

```
fid = fopen('resnet_accuracy_matlab.txt', 'w');  
fprintf(fid, 'ResNet-18 Accuracy: %.2f%\n', accuracy * 100);  
fclose(fid);  
  
fprintf('\n===== ResNet-18 Program Completed =====\n');
```

```
% Helper Function (Unchanged)
function [trainImgs, trainLbls, testImgs, testLbls] = loadCIFAR10Data(dataPath)
    trainImgs = [];
    trainLbls = [];
    for i = 1:5
        batch = load(fullfile(dataPath, sprintf('data_batch_%d.mat', i)));
        imgs = reshape(batch.data', 32, 32, 3, []);
        imgs = permute(imgs, [2 1 3 4]);
        trainImgs = cat(4, trainImgs, imgs);
        trainLbls = [trainLbls; batch.labels];
    end
    testBatch = load(fullfile(dataPath, 'test_batch.mat'));
    testImgs = reshape(testBatch.data', 32, 32, 3, []);
    testImgs = permute(testImgs, [2 1 3 4]);
    testLbls = testBatch.labels;
end

% This live script demonstrates how deep learning using convolutional tables (CTs)
% can be evaluated using a modified ResNet18 on CIFAR-10.
% The core idea is to use pretrained convolution layers and replace the
% dense layers with custom ones.
% This approach is also compatible with fewer resources.
```